HemoTwin

HAEMODIALYSIS DIGITAL TWIN

v1.0 · 2026 · research prototype

Predict the crash. Price the fix. Forecasts intradialytic hypotension 15–60 minutes ahead, then prices the ultrafiltration decision.

01 DATA

Machine telemetry
BP · UF · dialysate

HEMOBP — Taiwan
1,070 patients

205,482 sessions
4.19M telemetry rows

Mendeley — Spain
97,640 sessions · transport failed

Static EHR
age · sex · DM · vintage

02 FEATURES

Quality filters
session validity

Label builder
SBP<90 in (t+15, t+60]

40 causal features
shift(1) history only

Patient split
70 / 15 / 15, disjoint

Feature store
in-memory + manifest

03 MODELS

LightGBM
31 leaves · lr 0.02

Early stopping
on validation AUPRC

Isotonic calibration
held-out half of val

Event model
AUROC 0.916 · 8.4x lift

Rule baselines
R1 0.787 · R2 0.738

04 TWIN

Volume state
`dD/dt = UF - kD`

BP link
`SBP = SBP0 - beta*D`

Two parameters
k_refill · beta_bp

Counterfactual
hold · taper · stop

Residual band
90% interval, held-out

05 DASHBOARD

Chair board
8 chairs, ranked

Session view
BP · UF · forecast

Twin state
volume vs measured UF

What-if frontier
BP gain vs litres left

Trust panel
calibration · alarm budget

Model or simulation output

Clinical decision point

Pipeline component or artifact

Open data only: HEMOBP and Mendeley, CC BY 4.0. No PHI.